

EX. NO: 2b

Traveling Salesman Problem (Brute Force)

Date :04.08.2025

-

Aim:

To implement the Traveling Salesman Problem (TSP) using a brute force method in Python to find the shortest possible route that visits each city exactly once and returns to the origin city.

Procedure:

- 1. Understand the problem:**

The TSP requires finding the minimum-cost Hamiltonian circuit (a route visiting every node once and returning to start).

- 2. Represent the graph:**

Use a dictionary of dictionaries to represent the graph where keys are nodes and values are dictionaries of neighboring nodes with travel costs.

- 3. Generate permutations:**

Use Python's `itertools.permutations` to generate all possible sequences of visiting the nodes.

- 4. Calculate cost for each route:**

For each permutation, sum the travel costs between consecutive nodes plus the return trip to the starting node.

- 5. Track the minimum cost and path:**

Compare each permutation's total cost to find and store the minimum cost and corresponding path.

- 6. Output the result:**

Display the shortest path and its total cost.

Program Coding:

```
from itertools import permutations
```

```
def tsp_brute_force(graph):
```

```

nodes = list(graph.keys())

min_path = None

min_cost = float('inf')

for perm in permutations(nodes):

    cost = 0

    for i in range(len(perm) - 1):

        cost += graph[perm[i]][perm[i + 1]]

    cost += graph[perm[-1]][perm[0]] # return to start

    if cost < min_cost:

        min_cost = cost

        min_path = perm

return min_path + (min_path[0],), min_cost

# Example graph
graph = {

    'A': {'A': 0, 'B': 10, 'C': 15, 'D': 20},
    'B': {'A': 10, 'B': 0, 'C': 35, 'D': 25},
    'C': {'A': 15, 'B': 35, 'C': 0, 'D': 30},
    'D': {'A': 20, 'B': 25, 'C': 30, 'D': 0}

}path, cost = tsp_brute_force(graph)

print("Shortest path:", path)

print("Total cost:", cost)

```

Output:

❓ Given a distance matrix (graph), it tries **all permutations** of the cities.

❑ Returns the **shortest path and total distance**

Shortest path: ('A', 'B', 'D', 'C', 'A')

Total cost: 80

Result:

Thus, the Traveling Salesman Problem (TSP) using the brute force approach was successfully implemented.